Appendix E - Music Note and Frequency Tables

Each audio engine programs pitch through a divisor of one of two reference clocks. This appendix gives the divisor formula for each engine and a 12-note octave-4 table that programs can extend by halving the divisor for each octave up or doubling it for each octave down.

The reference frequencies are:

- **SoundChip / SFX / WAV / MOD / Amiga Paula DMA:** programmed in Hertz directly (the mixer runs at the system output sample rate). No divisor formula applies.
- **PSG / AY-3-8910:** master clock `2,000,000` Hz, period `clock / (16 * f)`.
- **SN76489:** master clock `3,579,545` Hz (NTSC), period `clock / (32 * f)`.
- **SID:** master clock `985,248` Hz (PAL), period `f * 16777216 / clock`.
- **TED audio:** master clock `894,886` Hz, period `clock / (8 * f) - 1`.
- **POKEY:** master clock `1,789,773` Hz (NTSC) divided by `28`, divisor `clock / 28 / f - 1`.

E.1 Octave 4 (middle C through B), equal temperament A4 = 440 Hz

Note	f (Hz)	PSG period	SN76489 period	SID period	TED period	POKEY divisor
C	261.63	478	427	4455	426	243
C#	277.18	451	403	4720	402	229
D	293.66	426	381	5001	379	216
D#	311.13	402	359	5298	358	204
E	329.63	379	339	5612	338	192
F	349.23	358	320	5947	319	182
F#	369.99	338	302	6300	301	171
G	392.00	319	285	6675	284	162
G#	415.30	301	269	7073	268	153
A	440.00	284	254	7493	253	144
A#	466.16	268	240	7939	239	136
B	493.88	253	226	8410	225	128

E.2 Extending the table

To move up one octave (double the frequency): halve the PSG / SN76489 / TED period, double the SID period, halve the POKEY divisor. The relation is exact to within the rounding error of the divisor: octave shifts of more than four become noticeably flat or sharp on the small-divisor chips (SN76489, POKEY) and need a tempered correction table for accurate music.

To move down one octave (halve the frequency): double the period (PSG, SN76489, TED), halve the SID period, double the POKEY divisor.

E.3 SoundChip and modern engines

The SoundChip channels and the sample-based engines (WAV, MOD, SFX, Amiga Paula DMA) take frequency in Hertz directly through the `FREQ` register of the channel block. A program writes `int(round(f))` into that register and the engine plays at that frequency. No table is needed: middle C is `262`, A4 is `440`.

E.4 Tuning notes

The numbers in section E.1 are calculated, not measured. The real-silicon SID, PSG, and POKEY chips drift with temperature and with the precise master-clock crystal; an Intuition Engine program that needs the same pitch on every machine should derive its own period from the engine clock at startup rather than hard- coding the values above.

The reference table assumes equal temperament. A program that wants just intonation, meantone, or other historical tunings generates the period table from the desired ratios using the same divisor formula. The engines have no awareness of a "scale": each channel plays whatever pitch its period register encodes.